

```

<!-- Code from d3-graph-gallery.com -->
<!DOCTYPE html>
<meta charset="utf-8">
<head>
<!-- Load d3.js and the geo projection plugin -->
<script src="https://d3js.org/d3.v4.js"></script>
<script src="https://d3js.org/d3-geo-projection.v2.min.js">
</script>

<!-- Load Leaflet -->

    <link rel="stylesheet"
href="https://unpkg.com/leaflet@1.9.3/dist/leaflet.css" />
    <script src="https://unpkg.com/leaflet@1.9.3/dist/leaflet.js">
</script>
    <!-- <script src="https://unpkg.com/leaflet@1.9.3/dist/leaflet.js">
</script> -->
    <link rel="stylesheet" href="./leaflet-mapwithlabels.css" />
    <script src="./leaflet-mapwithlabels.js"></script>
    <link rel="stylesheet" href="./leaflet.awesome-markers.css">
    <link href="http://netdna.bootstrapcdn.com/font-
awesome/4.0.0/css/font-awesome.css" rel="stylesheet">
    <link
href="http://netdna.bootstrapcdn.com/bootstrap/3.0.0/css/bootstra
rel="stylesheet">
    <link rel="stylesheet" href="https://unpkg.com/leaflet-control-
geocoder/dist/Control.Geocoder.css" />
    <script src="https://unpkg.com/leaflet-control-
geocoder/dist/Control.Geocoder.js"></script>
    <link
rel="stylesheet"
href="https://unpkg.com/leaflet-

```

geosearch@3.0.0/dist/geosearch.css"

/>

<script src="https://unpkg.com/leaflet-geosearch@latest/dist/bundle.min.js"></script>

<style>

body { margin: 0; padding: 0; }

#map_div { height: 600px; width: 100%; }

.info { padding: 6px 8px; font: 14px/16px Arial; background: white; border-radius: 5px; box-shadow: 0 0 15px rgba(0,0,0,0.2); }

.legend i { width: 18px; height: 18px; float: left; margin-right: 8px; opacity: 0.7; }

html { height: 100% }

body { height: 100%; margin: 0; padding: 0 }

#map-canvas { height: 100% }

html, body, #map-canvas { height: 100%; margin: 0; padding: 0; background: #f0f0f0; }

path.neighborhood {

cursor: pointer;

transition: fill-opacity 0.3s ease;

}

.leaflet-marker-icon, .leaflet-marker-shadow, .leaflet-image-layer,

.leaflet-pane > svg path, .leaflet-tile-container {

pointer-events: visible;

}

/* High-visibility label styling - EXTRA BOLD */

.d3-label {

font-family: "Helvetica Neue", Helvetica, Arial, sans-serif;

font-weight: 900;

fill: #111;

text-anchor: middle;

```
        pointer-events: none;
        text-shadow: 2px 2px 0px #fff, -2px -2px 0px #fff, 2px -2px
0px #fff, -2px 2px 0px #fff;
    }
    html, body, #map_div {
    height: 100% !important;
    width: 100% !important;
    margin: 0;
    padding: 0;
    -webkit-backface-visibility: hidden; /* Fixes "shimmer" on some
browsers */
}
```

```
/* Add this to your map page styles */
.leaflet-zoom-animated {
    display: block !important; /* Forces the SVG layer to stay
rendered */
}
```

```
.leaflet-tile-container {
    transition: opacity 0.2s linear; /* Makes tile loading less jarring */
}
```

```
/* Ensure the labels and polygons stay on the same rendering plane
*/
.leaflet-pane {
    z-index: 400;
}
```

```
/* Prevents the SVG renderer from clipping polygons during a resize
```

```

*/
/* Force the SVG container to always fill the available space */
/* Force the SVG container to fill the 100% space of the expanded
iframe */
.s{
  width: 100% !important;
  height: 100% !important;
  transform-origin: 0 0 !important;
  overflow: visible !important;
}

/* Ensure labels don't "stutter" during the move */
.leaflet-label {
  will-change: transform, top, left;
  transition: none !important;
}

/* Ensure the path (polygon) doesn't disappear or flicker */
/* Keep the polygon fill stable */
path.leaflet-interactive {
  /* Prevents borders from getting thick and overlapping during
zoom */
  vector-effect: non-scaling-stroke !important;
  stroke-width: 1px !important;
}

</style>
<link rel="stylesheet" href="./MarkerCluster.css" />
<link rel="stylesheet" href="./MarkerCluster.Default.css" />
<script src="./leaflet.markercluster-src.js"></script>
<!-- #SILAS <script src="./leaflet.markercluster.freezable.js">
</script> -->
<!-- After Leaflet script -->
<script
src="https://unpkg.com/leaflet.markercluster.freezable@1.0.0/dist/le

```

```
src="https://unpkg.com/leaflet.markercluster.freezable@1.0.0/dist/le
</script>
```

```
<!-- #SILAS This looks like the wrong file name to me -->
```

```
<!-- <script src="./LDeflate (1).js"></script> -->
```

```
<!-- #SILAS use the online version -- this is from the github
site -->
```

```
<script
```

```
src="https://unpkg.com/Leaflet.Deflate/dist/L.Deflate.js"></script>
```

```
<script src="./leaflet.awesome-markers.js"></script>
```

```
<!-- #SILAS I couldn't find the local version so I'm using the
online version below -->
```

```
<!-- <script src="./leaflet.markercluster.layersupport.js">
```

```
</script> -->
```

```
<!-- After Leaflet and Leaflet.markercluster scripts -->
```

```
<script
```

```
src="https://unpkg.com/leaflet.markercluster.layersupport@2.0.1/dis
</script>
```

```
</head>
```

```
<body>
```

```
<!-- Create an element where the map will take place -->
```

```
<div id="mapid"></div>
```

```
<style>
```

```
#mapid { height: 400px; }
```

```
</style>
```

```
<script>
```

```
// mapid is the id of the div where the map will appear
```

```

try {
  map = L.mapWithLabels('mapid', {
    center: [42.3840352, -71.1134377],
    zoom: 13,
    zoomSnap: 0,    // ALLOWS FRACTIONAL ZOOM (The D3
secret)
    zoomDelta: 0.1, // Smoother mouse wheel
    wheelPxPerZoomLevel: 120
  }); // center position + zoom
} catch (e) {
  console.warn("mapWithLabels plugin not found, using
standard Leaflet map.");
  map = L.map('map-canvas').setView([42.3840352,
-71.1134377], 13);
}

```

```

// Add a tile to the map = a background. Comes from
OpenStreetmap
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
  attribution: '&copy; OpenStreetMap contributors'
}).addTo(map);
// 1. Initialize Map

```

```

// Add a svg layer to the map
L.svg().addTo(map);

```

```

// Create data for circles:

```

```
var markers = [
  {long: 9.083, lat: 42.149}, // corsica
  {long: 7.26, lat: 43.71}, // nice
  {long: 2.349, lat: 48.864}, // Paris
  {long: -1.397, lat: 43.664}, // Hossegor
  {long: 3.075, lat: 50.640}, // Lille
  {long: -3.83, lat: 48}, // Morlaix
];
```

// Select the svg area and add circles:

```
d3.select("#mapid")
  .select("svg")
  .selectAll("myCircles")
  .data(markers)
  .enter()
  .append("circle")
    .attr("cx", function(d){ return map.latLngToLayerPoint([d.lat,
d.long]).x })
    .attr("cy", function(d){ return map.latLngToLayerPoint([d.lat,
d.long]).y })
    .attr("r", 14)
    .style("fill", "red")
    .attr("stroke", "red")
    .attr("stroke-width", 3)
    .attr("fill-opacity", .4)
```

// Function that update circle position if something change

```
function update() {
  d3.selectAll("circle")
    .attr("cx", function(d){ return map.latLngToLayerPoint([d.lat,
d.long]).x })
    .attr("cy", function(d){ return map.latLngToLayerPoint([d.lat,
d.long]).y })
```

```
}
```

```
// If the user change the map (zoom or drag), I update circle position:
```

```
map.on("moveend", update)
```

```
var neighborhoodData = [];
```

```
// 2. Define the D3 Overlay (Shapes + Labels + Interactions)
```

```
var d3Overlay = L.d3SvgOverlay(function (sel, proj) {
```

```
    // --- DRAW NEIGHBORHOOD SHAPES ---
```

```
    var paths = sel.selectAll('path').data(neighborhoodData);
```

```
    paths.enter()
```

```
        .append('path')
```

```
        .attr('class', 'neighborhood')
```

```
        .attr('stroke', 'black')
```

```
        .attr('fill', function() { return d3.hsl(Math.random() * 360, 0.7, 0.5); })
```

```
        .attr('fill-opacity', '0.5')
```

```
    // HOVER EFFECTS
```

```
    .on('mouseover', function(d) {
```

```
        d3.select(this)
```

```
            .transition().duration(150)
```

```
            .attr('fill-opacity', '0.8')
```

```
            .attr('stroke-width', 3 / proj.scale);
```

```
    })
```

```
    .on('mouseout', function(d) {
```

```
        d3.select(this)
```

```
            .transition().duration(150)
```



```

        .attr('fill-opacity', '0.5')
        .attr('stroke-width', 1 / proj.scale);
    })
    // CLICK TO ZOOM
    .on('click', function(d) {
        map.fitBounds(L.geoJson(d).getBounds());
    });

    // Sync path projection to map zoom/pan
    paths.attr('d', proj.pathFromGeojson)
        .attr('stroke-width', 1 / proj.scale);

    // --- DRAW LABELS ---
    var labels = sel.selectAll('text').data(neighborhoodData);

    labels.each(function(d) {
        // Find center of each polygon for label placement
        var center = L.geoJson(d).getBounds().getCenter();
        var point = proj.latLngToLayerPoint(center);
        d3.select(this)
            .attr('x', point.x)
            .attr('y', point.y)
            .style('font-size', (14 / proj.scale) + 'px');
    });
});

```

// 3. LOAD DATA AND ACTIVATE

// IMPORTANT: Ensure this filename matches your .geojson file exactly!

```

d3.json("./somervilleNeighborhoods (2).geo.json", function(error,
data) {
    if (error) {

```

```
    if (error) {  
        console.error("GeoJSON failed to load. Check file path or  
use a local server.", error);  
        return;  
    }  
  
    neighborhoodData = data.features;  
  
    // Add the D3 layer to the map now that we have data  
    d3Overlay.addTo(map);  
  
    console.log("Map initialized with", neighborhoodData.length,  
"neighborhoods.");  
  
});  
  
</script>  
</body>  
</html>
```